

DSA

Exercise 2:

E-commerce Platform Search Function

File:01Product.java

```
public class Product {  
  
    int productId;  
  
    String productName;  
  
    String category;  
  
  
    public Product(int productId, String productName, String category) {  
        this.productId = productId;  
        this.productName = productName;  
        this.category = category;  
    }  
  
  
    public void displayProduct() {  
        System.out.println("Product ID: " + productId);  
        System.out.println("Product Name: " + productName);  
        System.out.println("Category: " + category);  
    }  
}  
  
public class SearchDemo {  
  
  
    public static Product linearSearch(Product[] products, int searchId) {  
        for (int i = 0; i < products.length; i++) {  
            if (products[i].productId == searchId) {  
                return products[i];  
            }  
        }  
        return null;  
    }  
}
```

```

public static Product binarySearch(Product[] products, int searchId) {

    int left = 0;

    int right = products.length - 1;


    while (left <= right) {

        int mid = (left + right) / 2;


        if (products[mid].productId == searchId) {

            return products[mid];

        } else if (products[mid].productId < searchId) {

            left = mid + 1;

        } else {

            right = mid - 1;

        }

    }


    return null;

}

```

File 2: SearchDemo.java

```

public static void main(String[] args) {

    Product[] products = {

        new Product(101, "Laptop", "Electronics"),

        new Product(102, "Shoes", "Fashion"),

        new Product(103, "Mobile", "Electronics"),

        new Product(104, "Watch", "Accessories"),

        new Product(105, "Book", "Education")

    };


    int searchId = 104;

```

```
System.out.println("Linear Search Result:");
Product linearResult = linearSearch(products, searchId);

if (linearResult != null) {
    linearResult.displayProduct();
} else {
    System.out.println("Product not found");
}

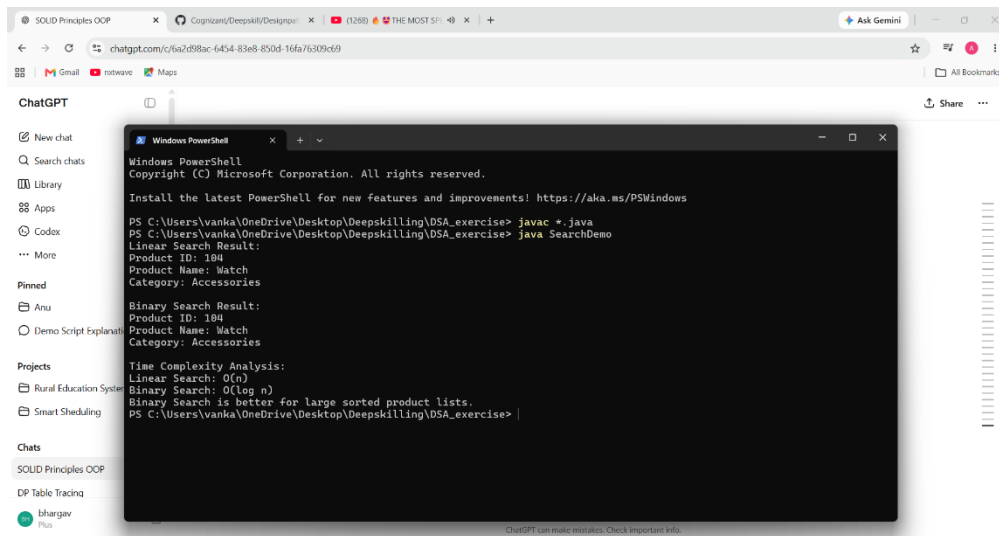
System.out.println();

System.out.println("Binary Search Result:");
Product binaryResult = binarySearch(products, searchId);

if (binaryResult != null) {
    binaryResult.displayProduct();
} else {
    System.out.println("Product not found");
}

System.out.println();

System.out.println("Time Complexity Analysis:");
System.out.println("Linear Search: O(n)");
System.out.println("Binary Search: O(log n)");
System.out.println("Binary Search is better for large sorted product lists.");
}
}
```



7.FinancialForecast

```
public class FinancialForecast {
```

```
    public static double calculateFutureValue(double currentValue, double growthRate, int years) {
```

```
        if (years == 0) {
```

```
            return currentValue;
```

```
        }
```

```
        return calculateFutureValue(currentValue * (1 + growthRate), growthRate, years - 1);
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        double currentValue = 10000;
```

```
        double growthRate = 0.10;
```

```
        int years = 5;
```

```
        double futureValue = calculateFutureValue(currentValue, growthRate, years);
```

```
System.out.println("Current Value: " + currentValue);
```

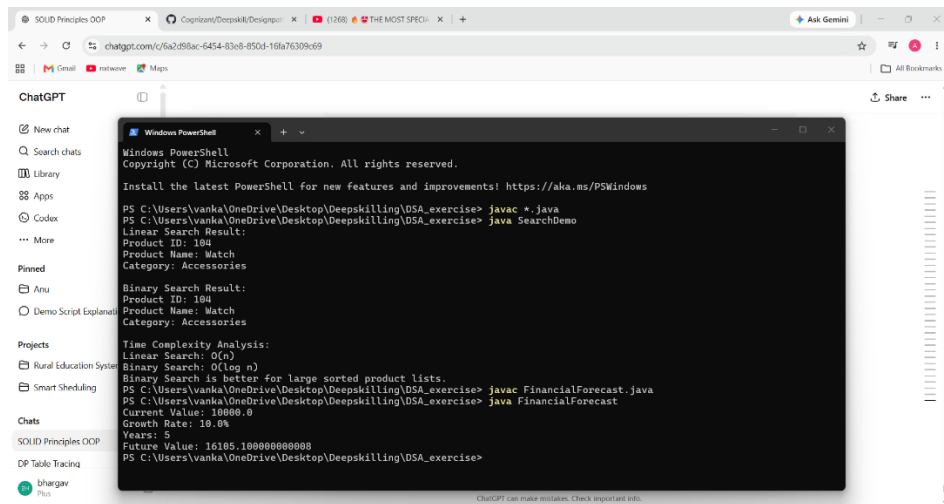
```
System.out.println("Growth Rate: " + (growthRate * 100) + "%");
```

```
System.out.println("Years: " + years);
```

```
System.out.println("Future Value: " + futureValue);
```

```
}
```

```
}
```



The screenshot shows a web browser window with a ChatGPT interface. A Windows PowerShell terminal window is overlaid on the ChatGPT interface, displaying the output of Java programs. The terminal output includes:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\vanka\OneDrive\Desktop\Deepskill\DSA_exercise> javac *.java
PS C:\Users\vanka\OneDrive\Desktop\Deepskill\DSA_exercise> java SearchDemo
Linear Search Result:
Product ID: 1004
Product Name: Watch
Category: Accessories

Binary Search Result:
Product ID: 1004
Product Name: Watch
Category: Accessories

Time Complexity Analysis:
Linear Search: O(n)
Binary Search: O(log n)
Binary Search is better for large sorted product lists.
PS C:\Users\vanka\OneDrive\Desktop\Deepskill\DSA_exercise> javac FinancialForecast.java
PS C:\Users\vanka\OneDrive\Desktop\Deepskill\DSA_exercise> java FinancialForecast
Current Value: 10000.0
Growth Rate: 10.0%
Years: 5
Future Value: 16105.100000000008
PS C:\Users\vanka\OneDrive\Desktop\Deepskill\DSA_exercise>
```